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Lesson 2-1: Quadratic Function**COMPLETE LESSON**

Student Name: _____ **Class / Group:** _____ **Date:** ____ / ____ / 2026 **Evaluation Score:** [____ / 10]

📖 Pre-Study: Check Your Readiness

Review the fundamental mathematical concepts required to master Quadratic Functions.

🔔 Prerequisite Review**1. Polynomial Degrees & Function Types**

The **degree** of a polynomial in one variable is the highest exponent of x with a non-zero coefficient:

- **Degree 0 (Constant Function):**
 $f(x) = c \rightarrow$ Graph is a horizontal line.
- **Degree 1 (Linear Function):** $f(x) = ax + b$ ($a \neq 0$) $\rightarrow$ Graph is an oblique straight line.
- **Degree 2 (Quadratic Function):**
 $f(x) = ax^2 + bx + c$ ($a \neq 0$) $\rightarrow$ Graph is a smooth curve called a **Parabola**.

2. Substitution & Negative Sign Mastery

Evaluating $f(k)$ means replacing every occurrence of x with the number k . Beware the sign rule:

- **Parentheses Squaring:** $(-3)^2 = (-3) \times (-3) = +9$
- **Negative Before Squared Term:**
 $-3^2 = -(3 \times 3) = -9$
- **Example:** For $f(x) = -x^2 + 5$, then
 $f(-2) = -(-2)^2 + 5 = -4 + 5 = 1$.

3. Expanding Binomial Squares (Perfect Squares)

Crucial for transitioning between *Vertex Form* and *Standard Form*:

- $$(x \pm h)^2 = x^2 \pm 2hx + h^2$$
- $(x - 3)^2 = x^2 - 6x + 9$
 - $(x + 2)^2 = (x - (-2))^2 = x^2 + 4x + 4 \rightarrow$ Note that $h = -2$!

4. The Cartesian Plane & Vertical Symmetry

Every vertical line has equation $x = h$. A parabola is perfectly symmetrical about its **axis of symmetry**:

- Points on the curve at equal vertical heights y are equidistant from the axis $x = h$. The midpoint between any two symmetric points gives the x -coordinate of the vertex!

☰ Diagnostic Check: Test Your Readiness

QUESTION 1 OF 3

1. What is the degree of the polynomial function $f(x) = x(x - 4) + 7$?

☐ A Degree 1 (Linear)

☐ B Degree 0 (Constant)

☐ C Degree 3 (Cubic)

☐ D Degree 2 (Quadratic)

QUESTION 2 OF 3

2. If $f(x) = 1 - x^2$, what is the exact value of $f(-3)$?

☐ A 10

☐ B 8

☐ C -8

☐ D -5

QUESTION 3 OF 3

3. Which expression is the algebraic expansion of $(x - 5)^2$?

☐ A $x^2 - 10x + 25$

☐ B $x^2 - 5x + 25$

☐ C $x^2 - 25$

☐ D $x^2 + 25$



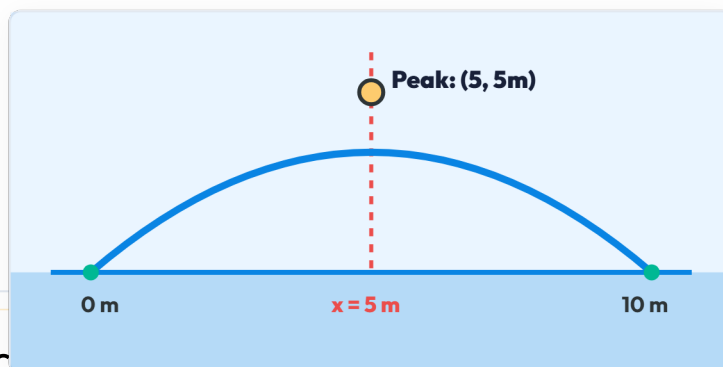
Real-World Connections (Mathematics in Everyday Life)

Precision Plots & Modeling

≡ Marine Acrobatics & Kinematics • Dolphin Leap Trajectory

🌐 STEM Real-World Application

In marine biology and physics, fluid dynamics models aquatic leaps using projectile parabolas. The curve $f(x) = -0.2x^2 + 2x$ maps horizontal distance against vertical clearance.



💡 Mathematical Concept

Kinematic Parabolic Modeling: $f(x) = -0.2x^2 + 2x$ • **Max Height** = 5 meters **at** $x = 5$ m .

The quadratic coefficient $a = -0.2 < 0$ produces a downward-opening parabola with a peak vertex at $x = -\frac{b}{2a} = -\frac{2}{2(-0.2)} = 5$ meters. Evaluating $f(5) = -0.2(25) + 2(5) = 5$ meters confirms that the dolphin attains its maximum altitude of 5 m precisely at the midpoint of its 10-meter leap.

Sports Science & Ballistics • Basketball Arc & Trajectory Analysis

STEM Real-World Application

Sports biomechanics models basketball shooting arcs through vertex form parabolas $f(x) = a(x - h)^2 + k$, revealing the precise mathematical conditions required to clear the rim.



Mathematical C

Vertex Form Trajectory Proof: ~~Test~~ $f(7) = -\frac{1}{3}(7 - 3)^2 + 6 = \frac{2}{3} \approx 0.67 \text{ m} \neq 3 \text{ m}$.

With vertex form $f(x) = -\frac{1}{3}(x - 3)^2 + 6$, the ball achieves its apex of 6 meters at $x = 3$ meters.

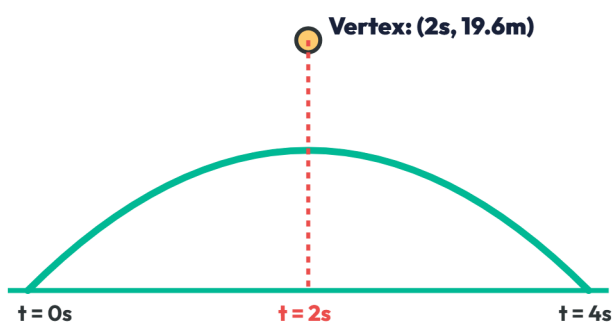
Evaluating the flight path at the basket distance $x = 7$ yields $f(7) = -\frac{1}{3}(16) + 6 = \frac{2}{3} \text{ m} \approx 0.67 \text{ m}$.

Because this height is well below the 3-meter hoop, the trajectory falls short of the target.

Kinematics & Physics • Football Projectile Peak & Flight Time

STEM Real-World Application

Gravitational acceleration ($g = 9.8 \text{ m/s}^2$) governs the symmetric flight of a kicked soccer ball, modeled by second-degree polynomial functions.



Mathematical Context

Gravitational Kinematics:

Peak Flight **Apex reached at** $t = 2$ seconds **with maximum altitude** 19.6 meters.
 For $h(t) = -4.9t^2 + 19.6t$, time to reach the apex is $t = -\frac{b}{2a} = -\frac{19.6}{2(-4.9)} = 2$ seconds. The ball ascends to a peak altitude of $h(2) = 19.6$ meters before symmetrically returning to earth at $t = 4$ seconds.

Essential Curriculum Standards

Definitions & Standard Forms

1

Definition & Domain of a Quadratic Function

A quadratic function is a second-degree polynomial function in one variable of the form $f(x) = ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$ and $a \neq 0$. Its domain is the set of all real numbers $\mathbb{R}$. Its geometric graph is a smooth, continuous U-shaped curve known as a **Parabola**.

2

Standard Form: $f(x) = ax^2 + bx + c$

- **x-coordinate of the Vertex:** $x = -\frac{b}{2a}$.
- **y-coordinate of the Vertex:** $y = f\left(-\frac{b}{2a}\right)$.
- **Axis of Symmetry:** Vertical line $x = -\frac{b}{2a}$.
- **y-intercept:** Point $(0, c)$ since $f(0) = c$.

3

Vertex Form: $f(x) = a(x - h)^2 + k$

- **Vertex Point:** Exactly (h, k) .
- **Axis of Symmetry:** Vertical line $x = h$.
- **Sign Alert:** For $f(x) = (x + 3)^2 + 1$, write it as $(x - (-3))^2 + 1$, giving vertex $(-3, 1)$.

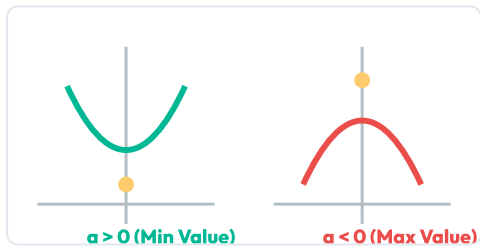
4

Orientation & Extrema (Maximum vs Minimum)

- If $a > 0$: Parabola **opens upward** (U); the vertex is a **Minimum Point**, with minimum value $y = k = f\left(-\frac{b}{2a}\right)$.
- If $a < 0$: Parabola **opens downward** (∩); the vertex is a **Maximum Point**, with maximum value $y = k = f\left(-\frac{b}{2a}\right)$.

**Encyclopedic Guide: Orientation, Vertex & Extrema****Geometric Analysis**

↕ Parabola Orientation & Vertex Extrema



- **When** $a > 0$: Parabola curves upward, having a valley at the minimum vertex.
- **When** $a < 0$: Parabola curves downward, having a peak at the maximum vertex.
- **Axis of Symmetry:** Divides the parabola into two congruent reflectional halves.

▲ Standard Form vs. Vertex Form Conversion

$$ax^2 + bx + c$$

$$a(x-h)^2 + k$$

$$h = -b / (2a)$$

$$k = c - b^2 / (4a)$$

- To find vertex from standard form: Compute $h = -\frac{b}{2a}$ then $f(h)$.
- To expand vertex form: $a(x - h)^2 + k = a(x^2 - 2hx + h^2) + k$.
- Both forms describe the exact same parabola with identical geometric properties.

Idea 1 • Standard

Concept Flashcard & Strategy Guide

Form

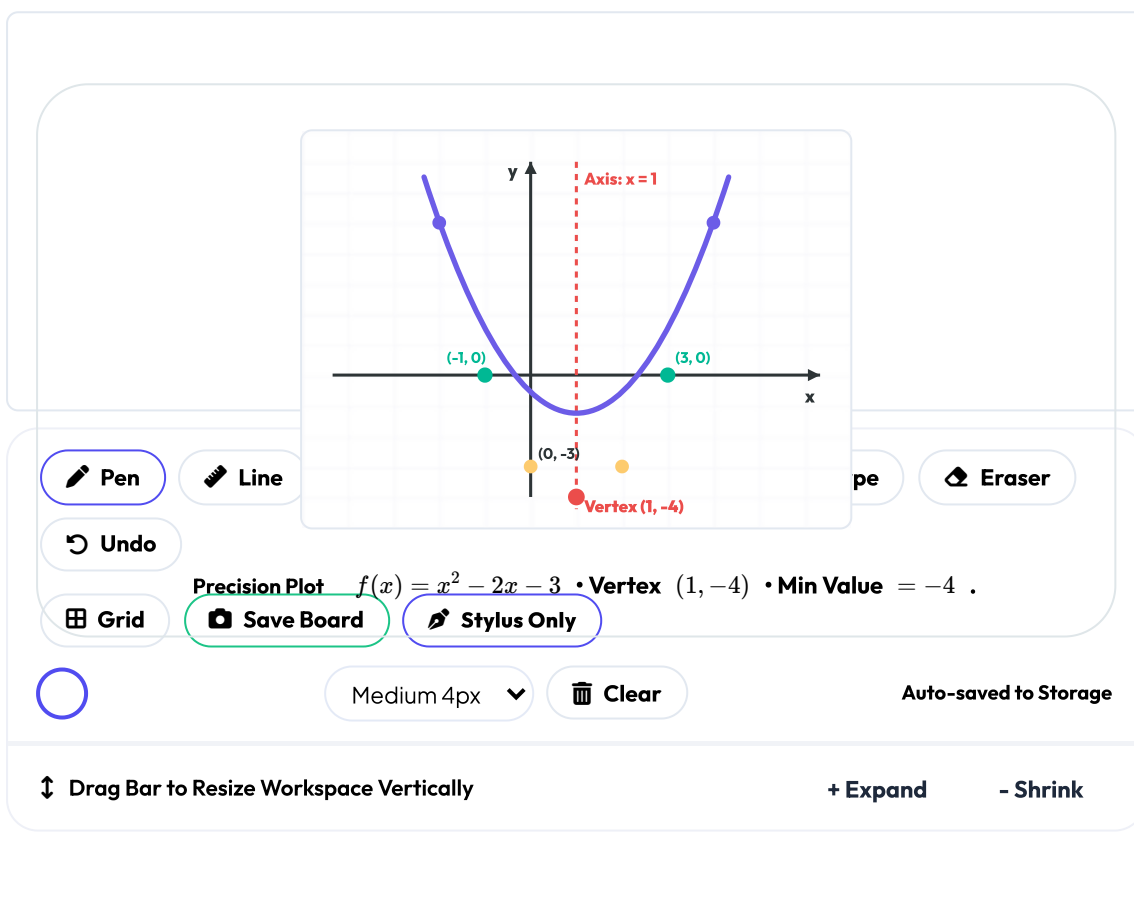
Graphing & Analyzing Standard Form: $f(x) = ax^2 + bx + c$

Follow the 4-step algorithm: (1) Find vertex $x = -\frac{b}{2a}$, (2) Select symmetric x -values around the vertex, (3) Construct the table of values, (4) Plot points and draw a smooth parabola.

Example 1.1 (Page 2)

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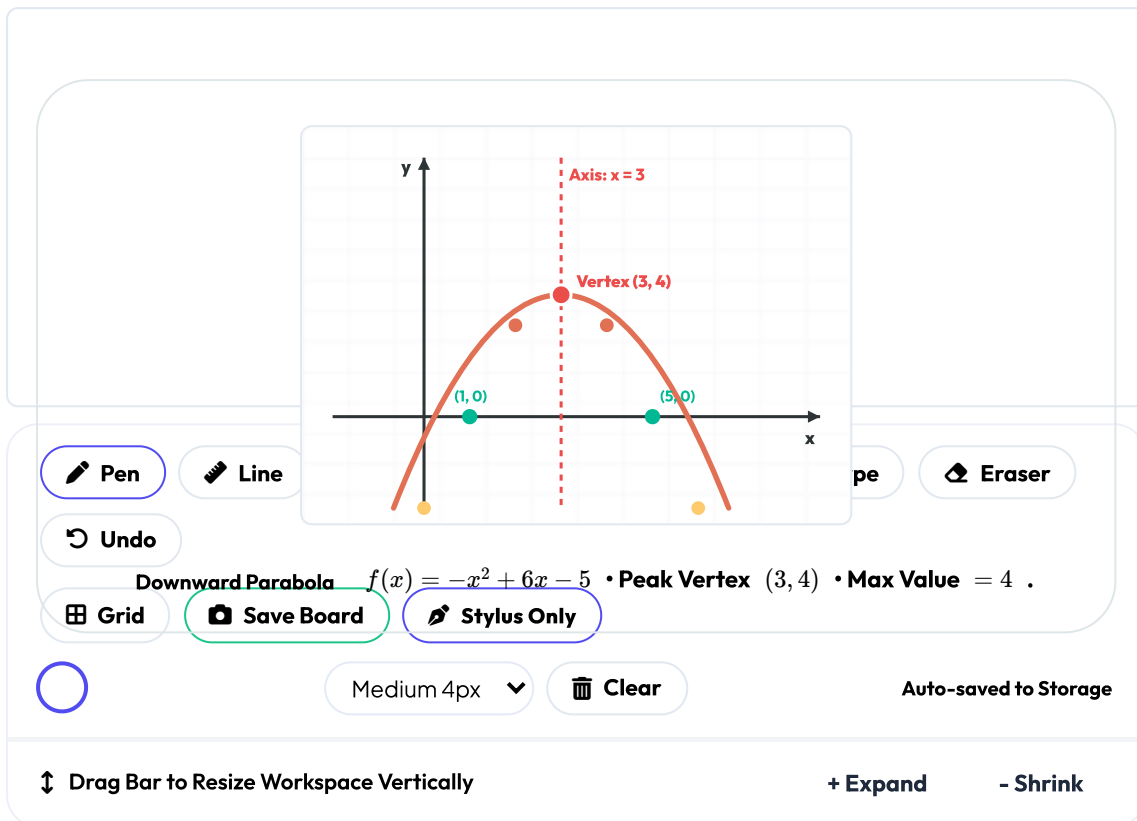
Graph the quadratic function $f(x) = x^2 - 2x - 3$. Identify the axis of symmetry equation and find the maximum or minimum value.



Example 1.2 (Downward Parabola)

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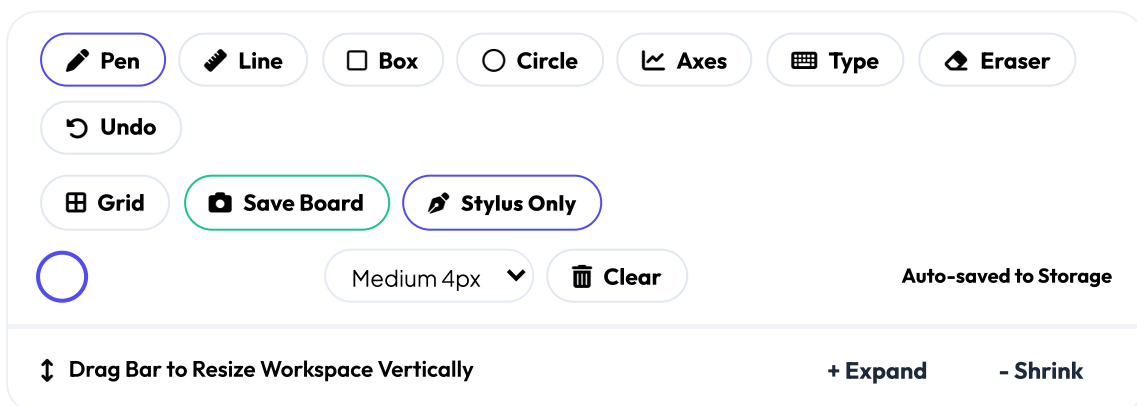
Graph $f(x) = -x^2 + 6x - 5$. State the vertex coordinates, axis of symmetry, and the maximum value.



Self-Assessment 1 (Page 2)

iPad Apple Pencil Workspace

Graph the function $f(x) = 3x - x^2$. On your drawing, clearly indicate the axis of symmetry and find the maximum or minimum value. Also verify whether the point (6, 21) lies on the curve.



Idea 2

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- **Vertex Form:** $f(x) = a(x - h)^2 + k$

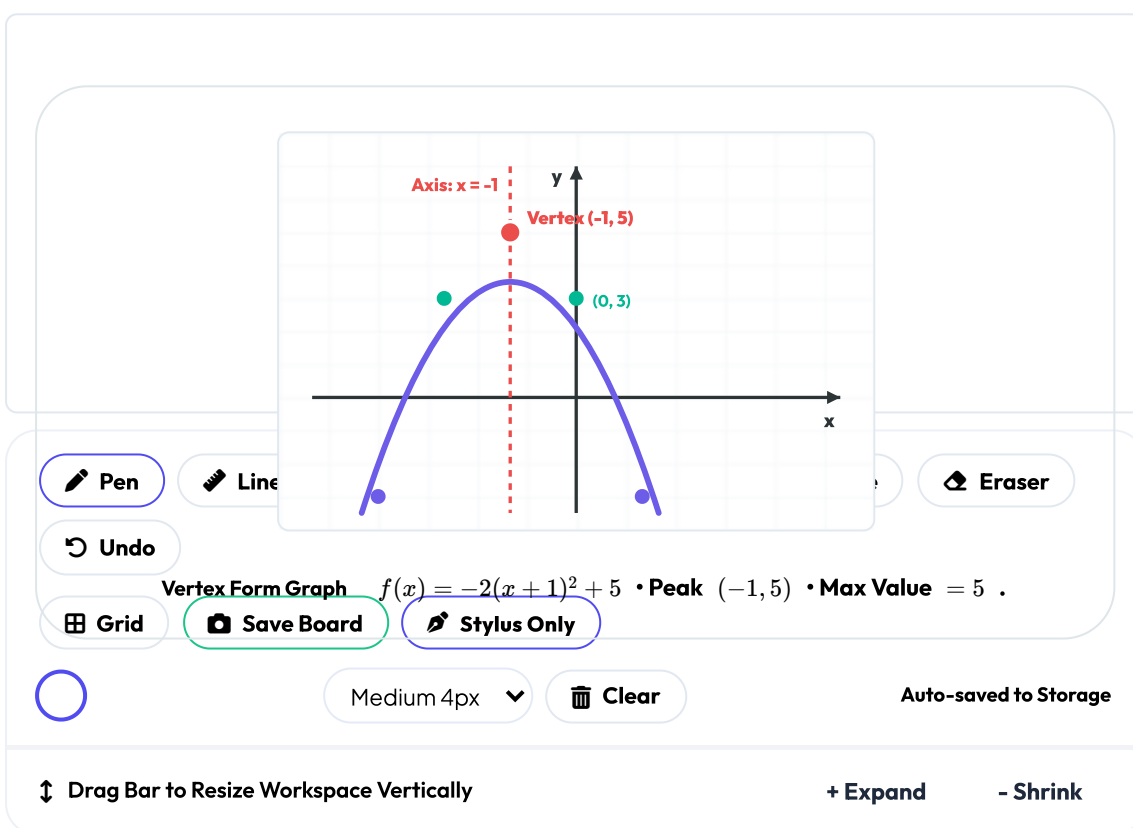
Vertex Form

The vertex is directly (h, k) without using formulas! Watch the negative sign inside: $(x - h)$ means h is positive, but $(x + 2)$ means $h = -2$. The vertical line $x = h$ is the axis of symmetry, and k is the maximum or minimum value.

Example 2.1 (Page 4)

iPad Apple Pencil Workspace

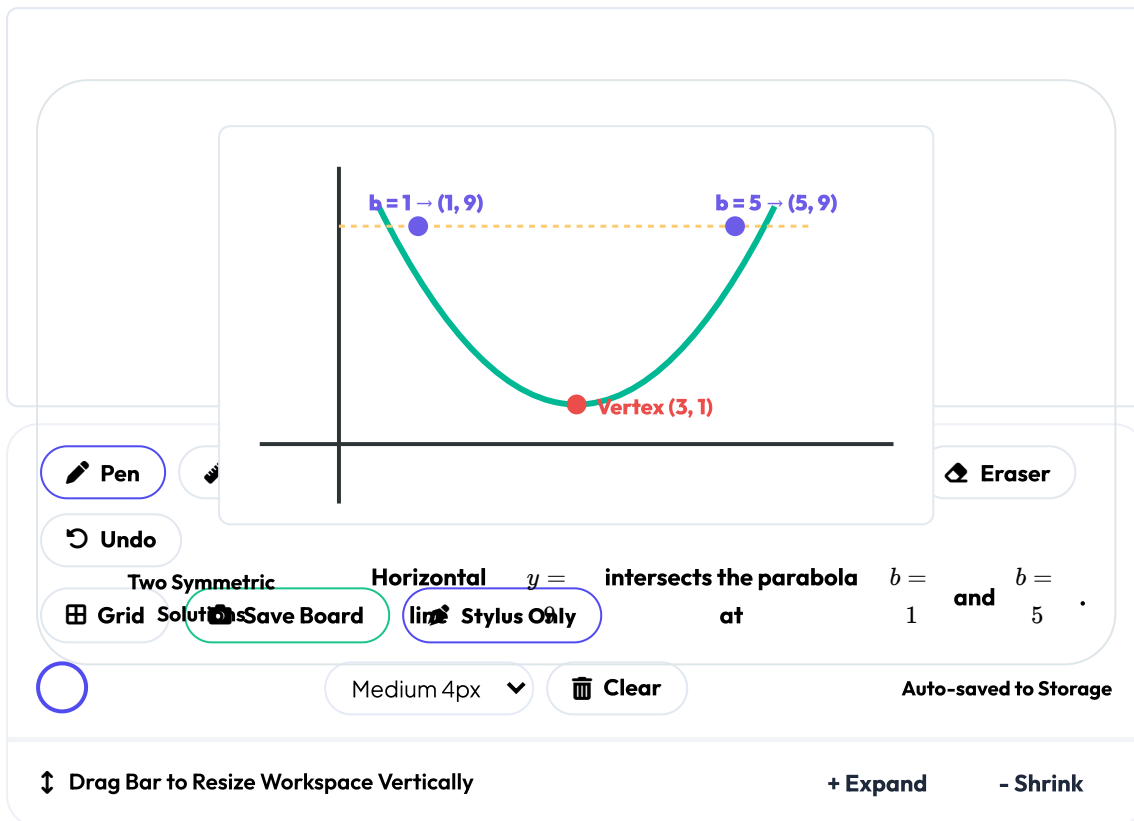
Graph $f(x) = 5 - 2(x + 1)^2$. State the vertex coordinates, axis of symmetry, and the maximum or minimum value.



 Example 2.2 (Page 5)

iPad Apple Pencil Workspace

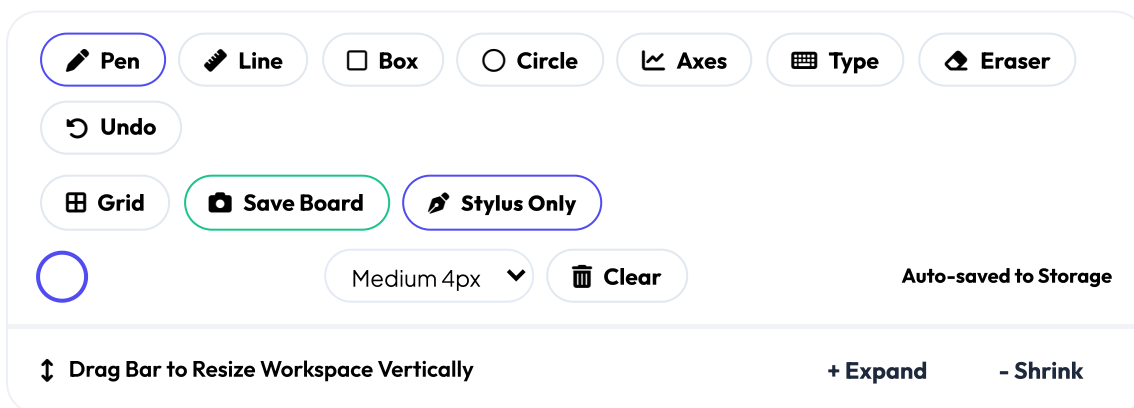
Given $f(x) = 2(x - 3)^2 + 1$. State the vertex and extrema. If the point $(b, 9)$ lies on the parabola, find all possible values of b .



 Self-Assessment 2 (Pages 4 & 5)

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- 1) Graph $f(x) = (x - 2)^2 - 4$. State the vertex, axis of symmetry, and minimum value.
- 2) If the point $(-1, m)$ lies on the curve of $g(x) = -(x + 1)^2 + 3$, calculate the value of m .



Idea 3 •

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Creative Solving for Unknown Parameters & Geometric Proofs

Problem

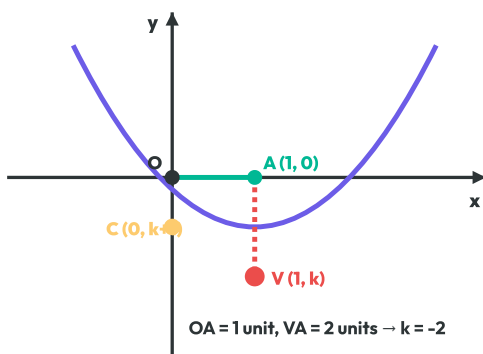
Use the conditions of the problem (e.g., degree of polynomial, coordinates of vertex, axis of symmetry, or geometric distance) to set up algebraic equations and solve for unknown constants.

Solving

Example 3.1 • Geometry & Parabola (Page 8 #12)

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In the figure, the curve represents $f(x) = x^2 - 2x + k + 1$ with vertex V and intersecting the y -axis at C . A is the projection of V on the x -axis ($x = 1$). If $OA = \frac{1}{2}VA$, find the value of k and the coordinates of point C .



Pen

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Eraser

Undo

Geometric

OA =

VA =

.

$$OA = \frac{1}{2}VA \Rightarrow 1 = \frac{1}{2}(-k) \Rightarrow k = -2$$

Grid

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Example 3.2 • Parameter & Degree Constraints (Page 7)

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- 1) If $f(x) = (a + 4)x^3 + x^2 - 5x + 6$ is a quadratic function, find a .
 2) If $x = 2$ is the axis of symmetry of $f(x) = ax^2 + 12x + 7$, find a .

Degree Constraint

Must be Degree 2

Coeff of $x^3 = 0$

$a + 4 = 0 \rightarrow a = -4$

Axis Constraint

Axis Formula: $-b / 2a$

$-12 / (2a) = 2$

$4a = -12 \rightarrow a = -3$

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High-Order Thinking (Page 8 #13)

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If $f(x) = x^2 - 10x + k$ is a quadratic function, and the vertex V is at a distance of 13 units from the origin $O(0, 0)$, find all possible values of the constant k .

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★ Self-Assessment Reflection

How confident do you feel with Lesson 2-1: Quadratic Function? Click your current level:


Need Practice


Getting It


Good Understanding


Mastered Confidently

Mr Ahmed Abd El-Motaal

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Empowering middle and high school students with engaging, interactive, visual mathematics. Dedicated to precision proofs, step-by-step clarity, and top exam preparation.

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